

Unit of analysis

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Topics to cover

- Review
- Unit of analysis
- Presentation 1: Jarvis & Hashimoto (2021)
- Presentation 2: Beinborn & Pinter (2023)

Review: Corpus design and selection

Corpus design and selection

- Sampling is important
- Representativeness
- Three-level framework
 - 1. Domain considerations
 - 2. Distributional considerations

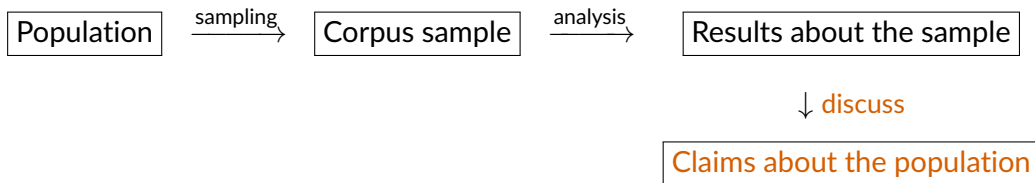
Corpus design and selection

- Sampling is important
- Representativeness
- Three-level framework
- Domain considerations
- Distributional considerations

Corpus design and selection

- Sampling is important

We observe the **sample**, but we often want to make claims about the broader **population**.



Corpus design and selection

- Sampling is important
- Representativeness
- Three-level framework
- Domain considerations
- Distributional considerations

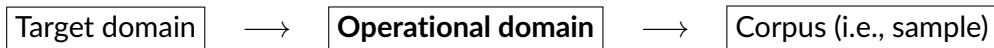
Corpus design and selection

- Sampling is important
- Representativeness is relative, but should be explicitly defined and described
- Three-level framework
- Domain considerations
- Distributional considerations

Corpus design and selection

- Sampling is important
- Representativeness is relative
- **Three-level framework**
- Domain considerations
- Distributional considerations

Corpus design and selection



- Target domain: the broader language-use domain
- **Operational domain: a concrete, researchable specification of that domain**
- Corpus: the actual texts

Corpus design and selection

- Sampling is important
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Corpus design and selection

Domain considerations

“...appropriate approximation of that target domain”

- Does the corpus represent the target language-use context?
- Are relevant L1 backgrounds adequately represented or balanced?
- Is proficiency reliably measured?
- Is annotation reliably done?

Distributional considerations

“...large enough and internally varied enough to reliable estimates of the linguistic variables of interest”

- Is the corpus large enough? (use CI to calculate the sample size)
- Are the target features sufficiently frequent?
- Are the target features well distributed across the texts?

Introduction



- Found/Compiled a corpus!



- How should I process my sample?



- How should I process my sample?
- In other words, what counts as **one observation**?
- Once we choose a unit of analysis, **how do we define and count it**?

What counts as one observation?

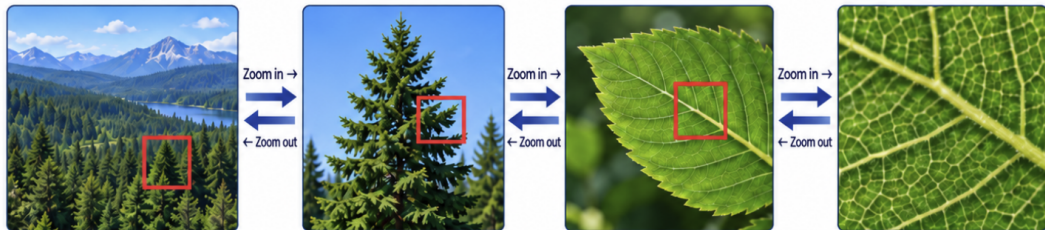
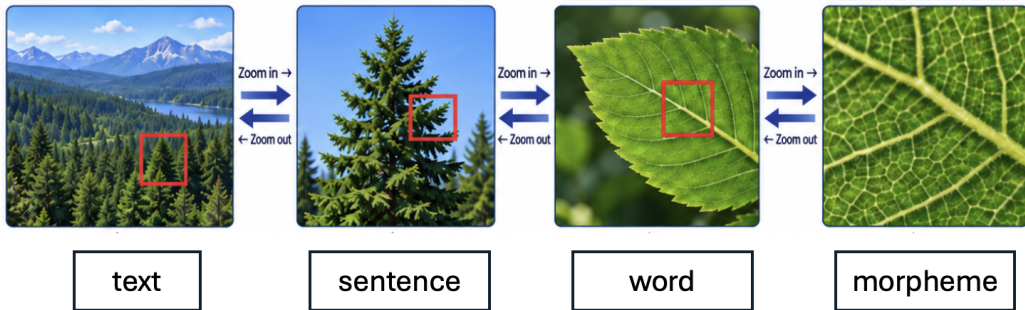
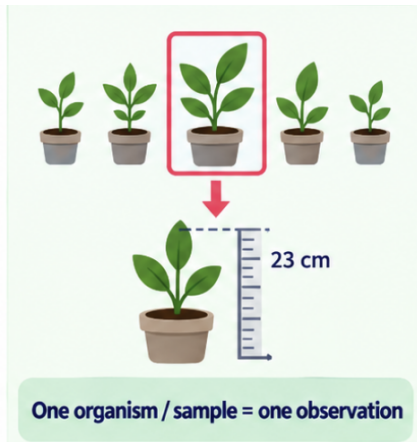
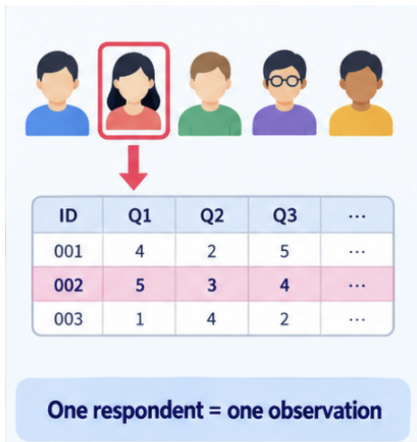


Image generated by ChatGPT

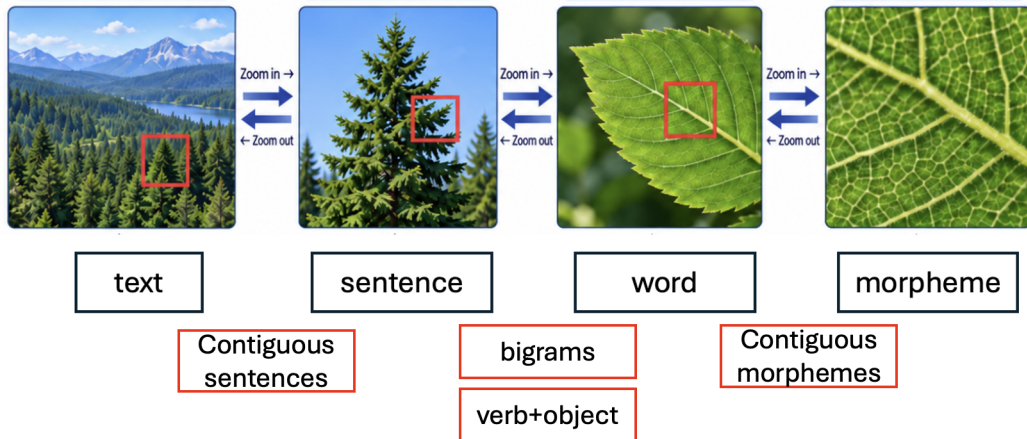
What counts as one observation?



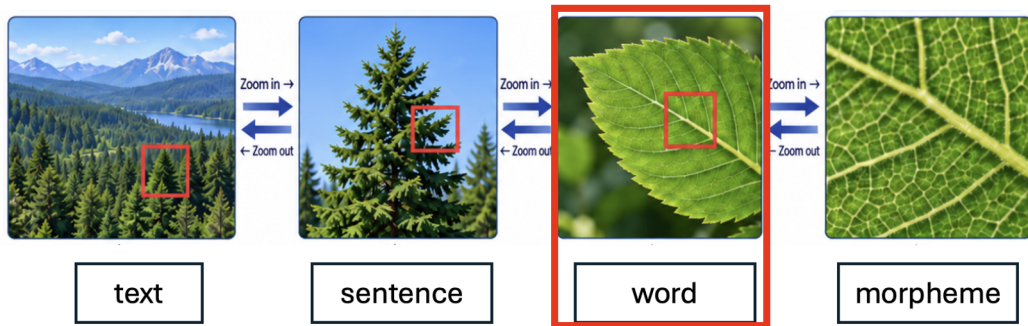
Other empirical studies?



Language data are nested



Word often treated as the basic unit of analysis



Words are often treated as a basic unit of analysis

But what exactly do we mean by a *word*?

- Is it always obvious where one word begins and another ends?
- For your study, what will count as one word?
- Once we decide this, can an NLP tool identify and extract words according to the definition?
- We will discuss based on English, but does “word” mean the same thing in other languages?

Lexical units

Two decisions when working with words

Two related but distinct decisions

1. Define the lexical unit

What should count as the same word?

Word type?

Lemma?

Flemma?

Word family?

2. Identify it computationally

**How can we find those units in
running text?**

Tokenization?

Lemmatization?

POS tagging?

Decision 1: What should count as the same word?

Consider the highlighted forms in context:

- She broke the school **record**. [N]
- The researcher **records** every interview. [V]
- The archive contains several historical **records**. [N]
- The interview was **recorded** yesterday. [V]
- They are **recording** the conversation now. [V]
- She bought a new digital **recorder**. [N]
- Several conversations remained **unrecorded**. [ADJ]

Decision 1: What should count as the same word?

How would you group these forms, and why?

- Should each form count as a different word?
- Should inflectional variants be grouped together?
- Should forms from different parts of speech be grouped together?
- Should derivationally related forms also belong to the same unit?

TASK:

- (1) Discuss your choice.
- (2) Compare it with the lexical units defined on the next slide.

Ways of operationalizing “word”, Webb (2021)

Unit	What is grouped together?
Orthographic form (word type)	Each distinct surface form is treated as a separate unit.
Lemma	A headword and its inflectional forms within the same part of speech.
Flemma	A headword and its inflectional forms, without distinguishing part of speech.
Word family	A headword together with its inflectional and derivationally related forms.

The choice of lexical unit matters, Webb (2021)

Changing the lexical unit changes what we count.

Smaller units

- Treat more forms separately
- Require fewer assumptions about morphological knowledge
- Can distinguish knowledge of different forms more precisely

Larger units

- Group morphologically related forms
- Capture relationships among related words
- Require stronger assumptions about what learners know

The choice can affect estimates of: (1) vocabulary size, (2) lexical diversity, (3) lexical coverage (i.e., the proportion of running words in a text covered by known vocabulary)

The choice of lexical unit matters, Webb (2021)

- Learner proficiency
Can we assume that knowing one form implies knowledge of related inflected or derived forms?
- Receptive vs. productive knowledge
Recognizing a related form does not necessarily mean that a learner can produce it accurately.
- Conventions in the specific research area
What unit is typically used for the construct we are measuring? (e.g., lexical diversity)

Decision 2: How do we find words in running text?

Once we decide what kind of lexical unit we want, we still need to identify units in actual corpus data. Begins with **tokenization**.

Tokenization

The process of segmenting continuous input into a sequence of “computational” units.

The students recorded the interview.

may be tokenized as:

The | students | recorded | the | interview | .

▶ Token is NOT type!

Identify it computationally

These tokens can then be used for POS tagging, lemmatization, parsing, and other analyses.

Target unit	Possible computational requirements
Orthographic form (word type)	Tokenization + decisions about normalization (e.g., capitalization, punctuation)
Lemma	Tokenization +
Flemma	Tokenization +
Word family	Tokenization +

Even tokenization involves decisions

What should a tokenizer do with these?

I'm well-known 3.14 U.S. ...

Should I'm be:

I'm **or** I | 'm ?

And in spoken corpora:

um uh repetitions false starts word fragments

Whether these count as relevant tokens depends on **the purpose of the analysis.**

Word boundaries are not universal

What happens if we identify words simply by **splitting on whitespace**?

Chinese / Japanese / Thai

- Spaces do not consistently correspond to word boundaries.
- In Chinese and Japanese, a lexical word may consist of multiple characters.

Korean

- Spaces generally mark *eojeol* boundaries.
- One *eojeol* may contain a lexical stem together with one or more grammatical morphemes.
- An *eojeol* is therefore not necessarily equivalent to a lexical word.

Splitting on spaces may be useful for some languages and purposes, but it is not a language-independent method for identifying words.

Should we use smaller units instead?

If “word” is difficult to define, we can consider smaller units.

Morpheme

A meaningful subpart of a word

long + -er

Character

A clearly identifiable computational unit

A, 한, 木

Subword

A smaller sequence learned or defined for computational processing

low | er

But smaller units introduce their own trade-offs:

- Morpheme boundaries are not equally clear across languages.
- Characters are computationally convenient but may be linguistically too small.
- Subword units need not correspond to meaningful morphemes.

Subword tokenization

Modern NLP systems often avoid relying exclusively on whole-word vocabularies.

Suppose a system knows:

low new newer

but has not encountered:

lower

A subword tokenizer may represent it as:

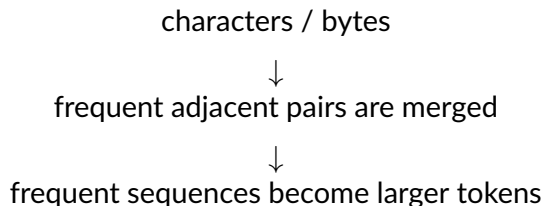
low | er

Important

A segmentation such as low | er does NOT necessarily mean that the tokenizer has identified a root and suffix. The segmentation may simply reflect frequently occurring character sequences in the training data.

Byte-Pair Encoding (BPE): Basic idea

BPE learns subword units from frequency patterns.



e.g.,:

$l + o \rightarrow lo \quad \rightarrow \quad lo + w \rightarrow low$

Frequently occurring forms may become whole tokens, whereas rarer forms remain segmented into smaller pieces. In short, BPE learns statistical units, not linguistic analyses (Demo: <https://tiktokenizer.vercel.app/>)

Takeaway

“Word” is not simply something that a computer finds for us.

- We first decide what linguistic unit is relevant to the research question.
- We then give that unit an operational definition.
- Only then can we decide how to identify that unit computationally.

Jarvis & Hashimoto (2021)

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- Examines how the choice of lexical unit affects measures of lexical diversity.
- Shows that lexical diversity estimates can change depending on how words are operationalized.

Beinborn & Pinter (2023)

Beinborn & Pinter (2023)

- Evaluates whether common subword tokenizers (including BPE) produce segmentations that are cognitively plausible.
- Compares tokenizer behavior with lexical decision data across several languages.

Appendix

Tokens are not the same as types

Consider:

The student recorded the interview.

If capitalization is ignored and punctuation is counted:

the | student | recorded | the | interview | .

Tokens

Individual occurrences in the text

$$N = 6$$

Types

Distinct forms among those tokens

$$|V| = 5$$

The word *the* occurs twice, but represents one type.